

QANDEEL

قندیل

Herd, don't harvest: autonomous, live-release management of Blue Blubber jellyfish swarms after ENEC's early warning

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We keep swarms out of the intake and return them to the sea alive

Problem

Blue Blubber swarms (30–45 cm, 1–3 kg) drift onto seawater intakes. Clearing them takes heavy manual work and almost none survive.

Solution

After ENEC’s alert, a drift planner decides when to act; a bubble curtain at the breakwater gap holds the swarm out; two uncrewed boats tow a boom with a closed-bottom bag and release the jellyfish alive. An AI gap camera is the backup.

Evidence

Simulation on real 2025 Gulf currents and wind, stress-tested against physics the planner does not know; online dashboard; 19 tests; open code.

Next

A 1:20 tank test (AED 300 kit) to measure what the model assumes, then a sea pilot with ENEC.

0.1% vs 17%

of arriving jellyfish reach the gap before the curtain runs: Qandeel vs a daylight camera alone (stress test, 100 alerts, real 2025 data)

1–2 days

of warning for the boat crews and the curtain, for about USD 170 of extra energy per alert

84–100%

of jellyfish kept by the boom with our retention bag (model), vs 0–55% for an open oil-spill boom

46%

of the energy of a fixed airflow, by setting air hour by hour from real currents and wind

A large, local, hard-to-handle jellyfish

ENEC already detects swarms offshore. The open question is what to do next: manage high volumes with less manual work and less harm.

The target: Blue Blubber

- One of the two dominant outbreak species on the UAE coast; reproduces year-round (Aravind 2024)
- 30–45 cm bell, ~1–3 kg each (Yang et al. 2018 mass formula)
- 87–100% sit in the top 1 m by day (Pitt & Kingsford)
- Large adults swim >10 cm/s: not just passive drifters

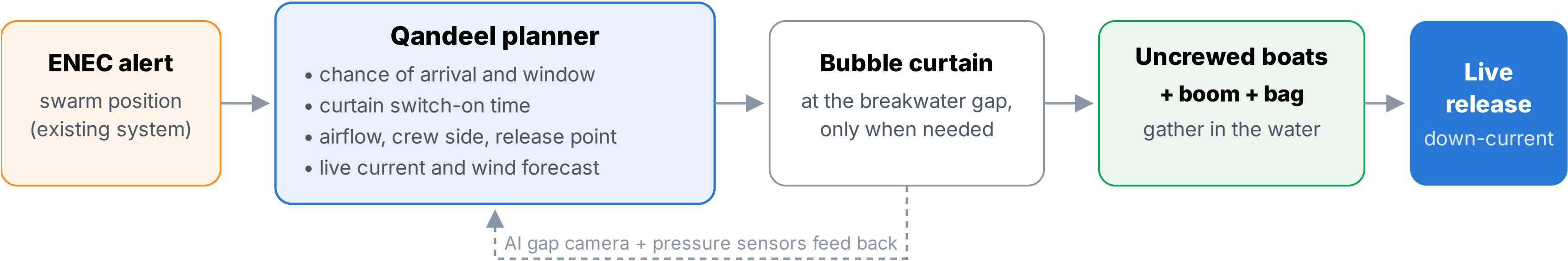
What happens at intakes

- Swarms follow the intake flow through the breakwater opening and pile on screens
- Worldwide this has forced power reductions and shutdowns (e.g. Torness 2011, Oskarshamn 2013)
- In the region, ~300 t of jellyfish clogged intakes in Oman

Why it is hard

- Manual raking and trucking, sometimes for days
- Jellyfish crushed on screens or discarded: almost none survive
- Cutting robots (e.g. JEROS) shred them into fragments
- No management method has ever been tested on this species

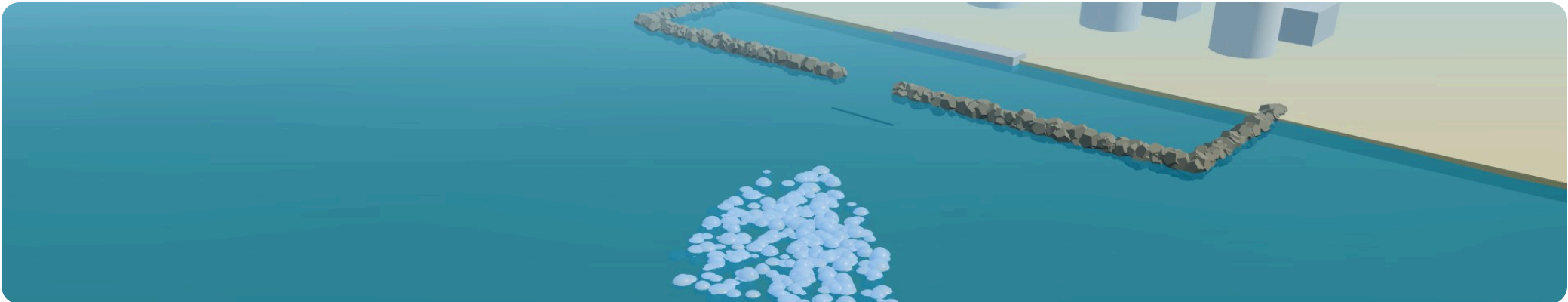
Guard the one gap, move the water, keep every jellyfish in the sea



1 • The gap is a chokepoint
Seawater, and the swarm, can only reach the intake through one opening.

2 • Move the water, not the animal
A rising-bubble current holds them out; nothing is cut, pumped or netted.

3 • Proven tools, made autonomous
Oil-spill booms towed by uncrewed boats; with a bag that stops jellyfish leaking.



The breakwater leaves one opening to the intake. A swarm offshore must pass through it, so that is the one place we act. Image: our 3D model (stylised; not to scale).

One swarm, from alert to live release

- 1

Alert.

ENEC reports a swarm, e.g. 15 km NNW. The planner receives it automatically.
- 2

Plan.

400 simulated drifts of 60 swimming jellyfish each, on the live current and wind forecast: 54% chance of arrival, window 10–77 h.
- 3

Prepare.

Curtain on at +7 h; uncrewed boats staged on the approach side; release point chosen.
- 4

Deflect.

Curtain across the gap holds the swarm outside; airflow follows the current.
- 5

Gather.

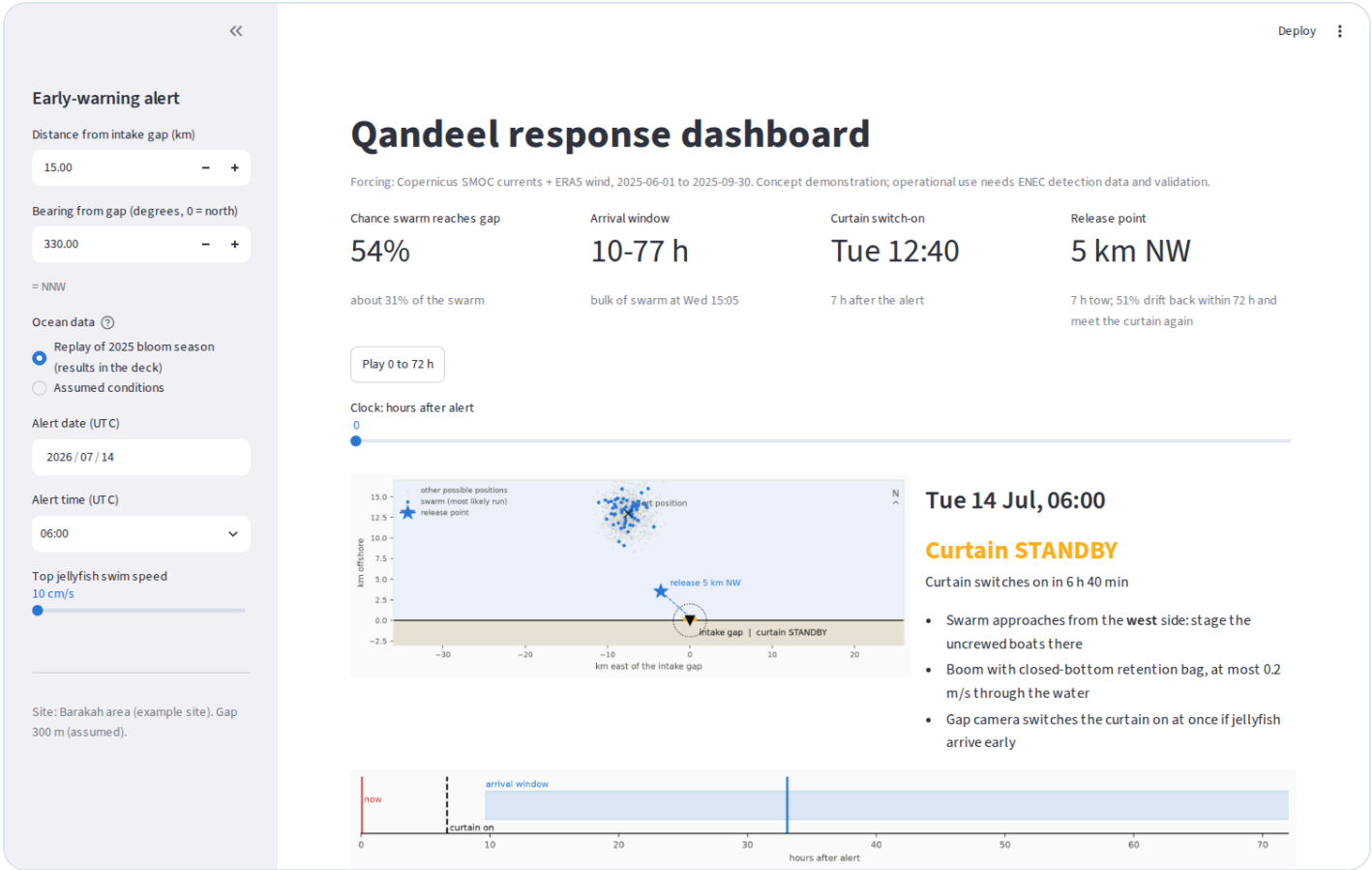
Two uncrewed surface vessels, supervised from shore, close a U-boom with a closed-bottom bag.
- 6

Release.

Tow ~5 km at ≤ 0.2 m/s (about 7 h) and open the bag; about half drift back and are held again by the curtain.
- 7

Learn.

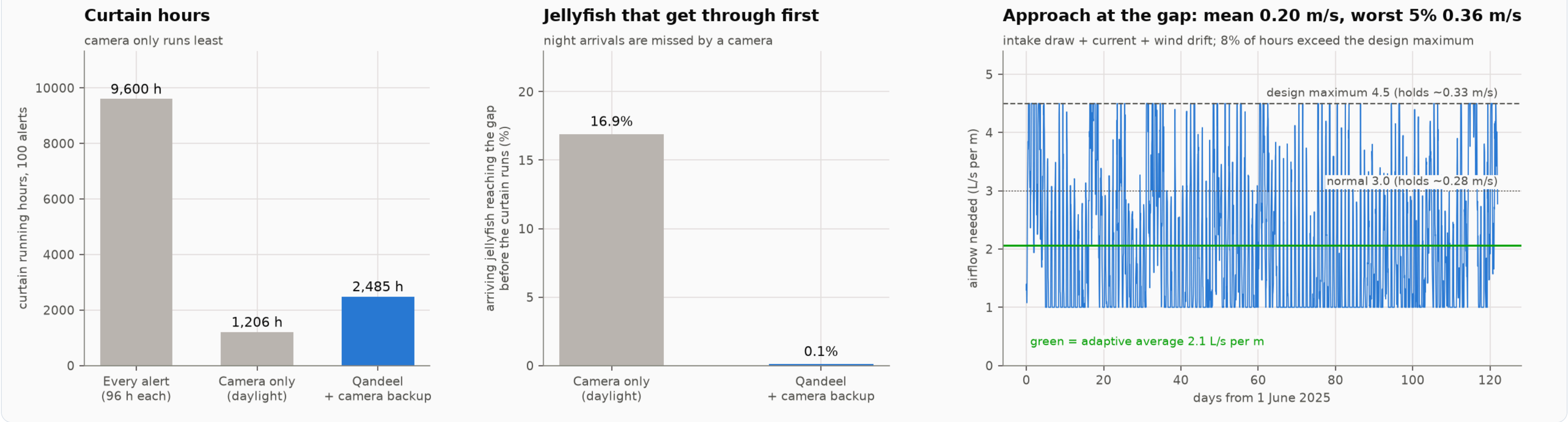
An AI camera counts jellyfish at the gap and starts the curtain if they arrive early. Its records are the training data for a learning model that corrects the drift forecast.



Qandeel operator dashboard (working prototype, Streamlit): chance of arrival, window, switch-on clock time, swarm map, curtain status, release point. Data: Copernicus SMOC currents + ERA5 wind, summer 2025.

Better protection than a camera alone, at a known cost

Model-error stress test, 100 alerts in the real 2025 season: Qandeel lets 0.1% through vs 17% for a camera alone



100 alerts at random times and places in the real 2025 season; the truth uses physics the planner does not know. "Arrived" = within 2 km of the gap (15 km NNW alert: 54%; at 500 m, 48%).

0.1% vs 17% get through

Truth drifts with currents $\times 0.7\text{--}1.3$, wind drift 1.5–5% and an unmodelled 0–3 cm/s current. A daylight camera alone misses night arrivals; Qandeel switches on ahead of them.

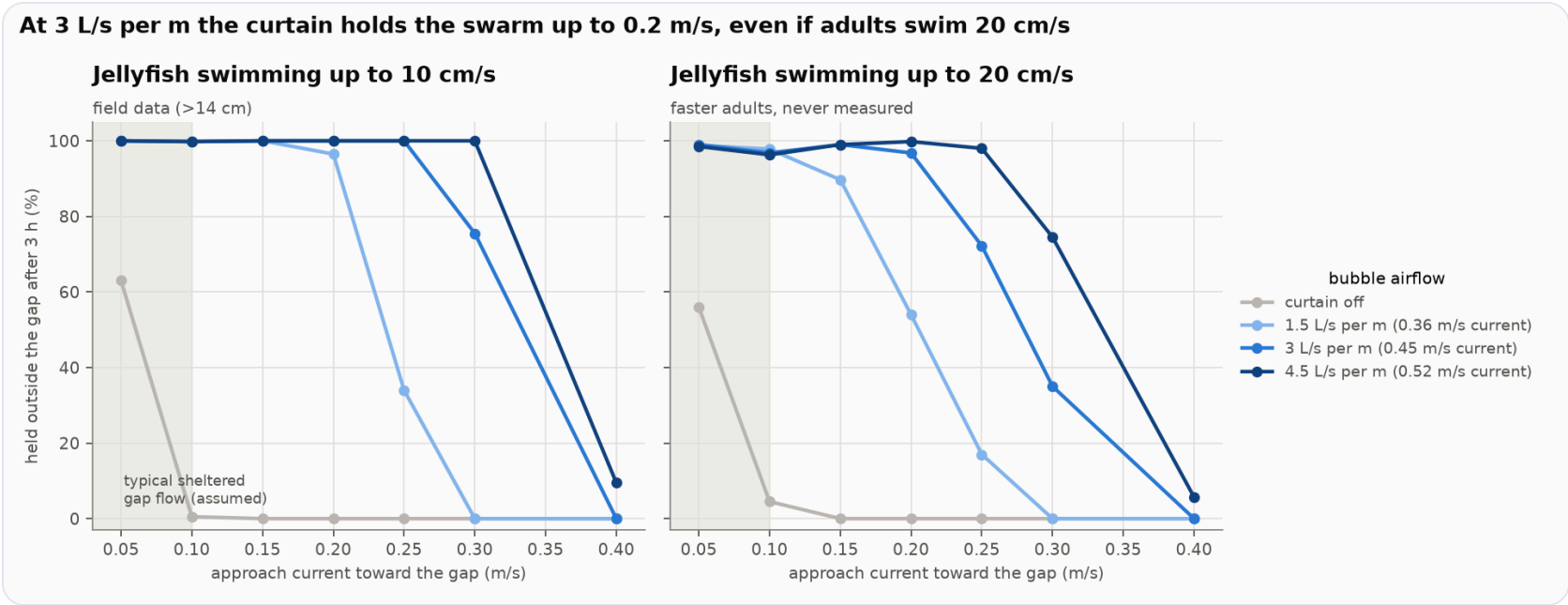
The cost

2,485 curtain hours vs 1,206 for camera-only (about 13 h, USD 170 of energy, per alert), and 74% fewer than running on every alert. In return: 1–2 days of warning.

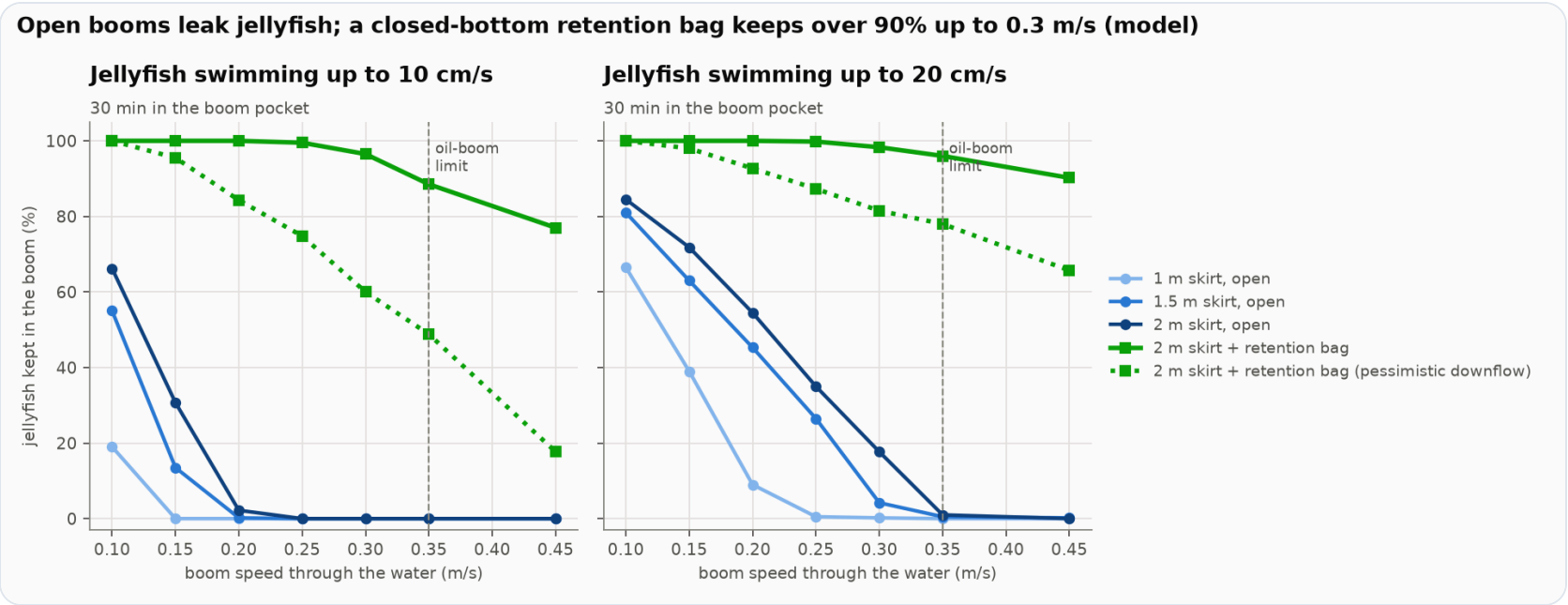
What this is not

Still simulation against simulation, not real jellyfish. Next: compare drift with real Gulf drifter tracks and ENEC swarm records.

Our finding: oil-spill booms leak jellyfish, and a bag fixes it



400 individually simulated jellyfish per point; surface current from Bulson's pneumatic-barrier relation.



Jellyfish depth from field data; apex downflow 25% (pessimistic 50%) of tow speed.

Curtain holds the swarm, with little spare

Water approaches the gap at ~0.20 m/s on average (intake draw + current + wind), 0.36 m/s in the worst 5% of hours. Normal setting 3 L/s per m holds ~0.27 m/s; design maximum 4.5 holds ~0.33 m/s. ~8% of hours exceed it and are flagged to operators.

Why the bag matters

Oil floats back up; neutrally buoyant jellyfish are dragged under the skirt far below the 0.35 m/s oil-boom limit (0–55% kept at 0.2 m/s). A closed-bottom fabric bag keeps 84–100% in the model.

Gentle on marine life, built from parts that exist today

Marine-life protection

RISK	HOW WE REDUCE IT
Bubbles drawn into the intake	Curtain on the seaward side; bubbles surface in under a minute, in the outer basin; bubble detector on the intake line; curtain moved further out if the pilot shows carry-over
Nuclear approval, security	FANR approval from day one; kit stays outside safety-related intake structures; boats operated by plant staff inside the security zone rules
Jellyfish harmed	No cutting, pumping or mesh; coarse bubbles push them away at the surface (air under the bell checked in the tank test)
Turtles, dolphins, seabed	Escape gap, slow tow, camera stop on sighting; pipe on legs above the bed, away from coral and seagrass
Released jellyfish strand	Left alone, 57% of the same swarm strands within 4 days; release ~5 km out strands 40%
Carbon	~11 MWh per event: ~4.4 t CO ₂ on a gas grid, ~0.1 t on the plant's own power

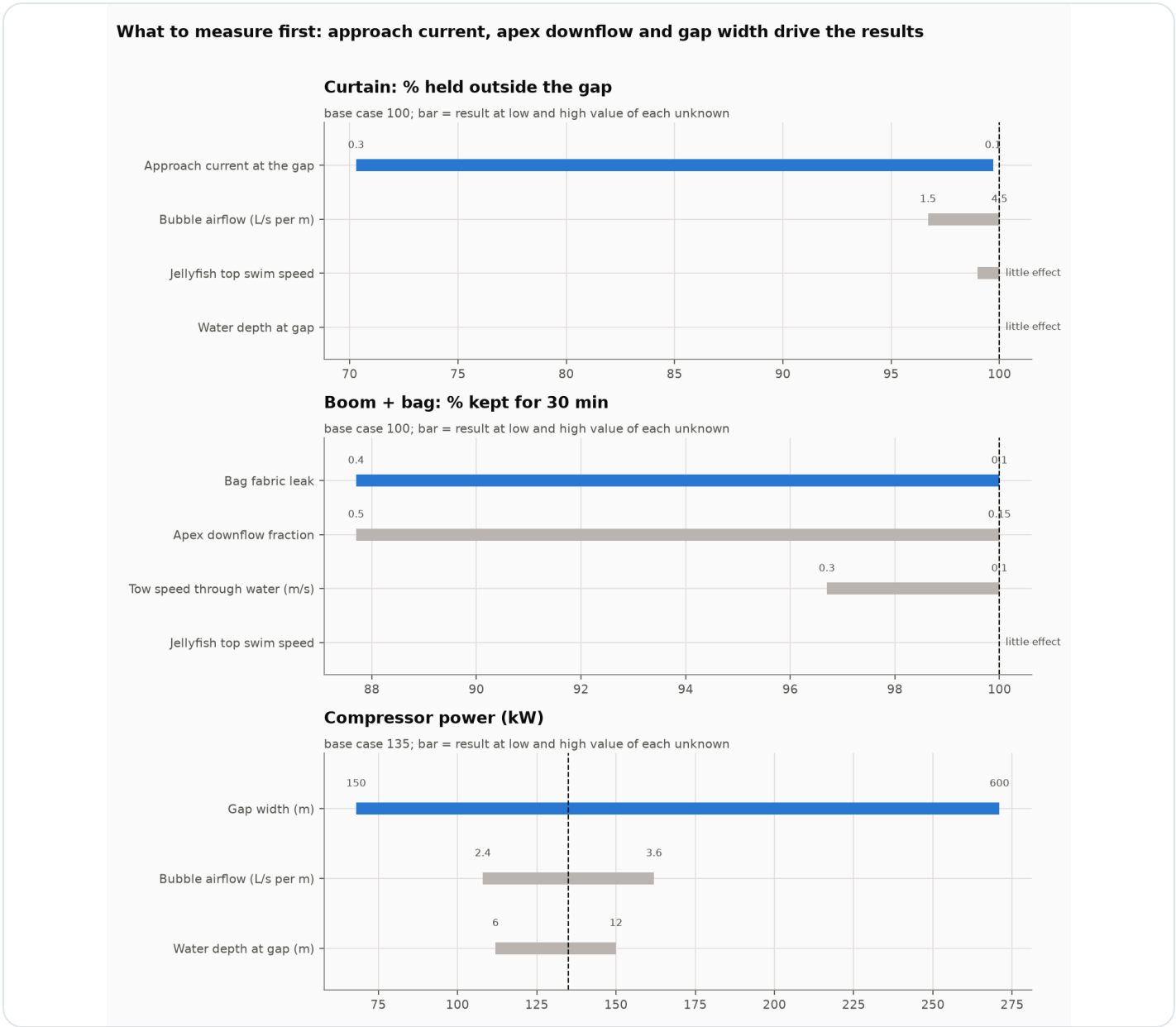
Feasibility

ITEM	ESTIMATE (300 M GAP)
Compressed air	0.9 m ³ /s, ~135 kW, oil-free (ISO 8573-1 Class 0)
Capital (indicative)	USD 0.55–1.8 M: curtain, compressor + standby, boom, bags, two uncrewed boats, sensors
Running cost	~USD 1,100 of energy per event
Value	One to four avoided 12 h unit outages (~USD 0.5 M each, EPRI figure) pay it back
Capacity	Gathering 4–43 t/h, but the loaded tow sets the pace: ~0.5 t/h net per boom unit, ~1.8 t/h with three uncrewed tugs swapping bags. The curtain does the bulk work; boats clear what builds up
Gulf hardening	Membrane diffusers, foul-release coating, modular sections, lifted out off-season

What we assume, what we don't know yet, and what we measure first

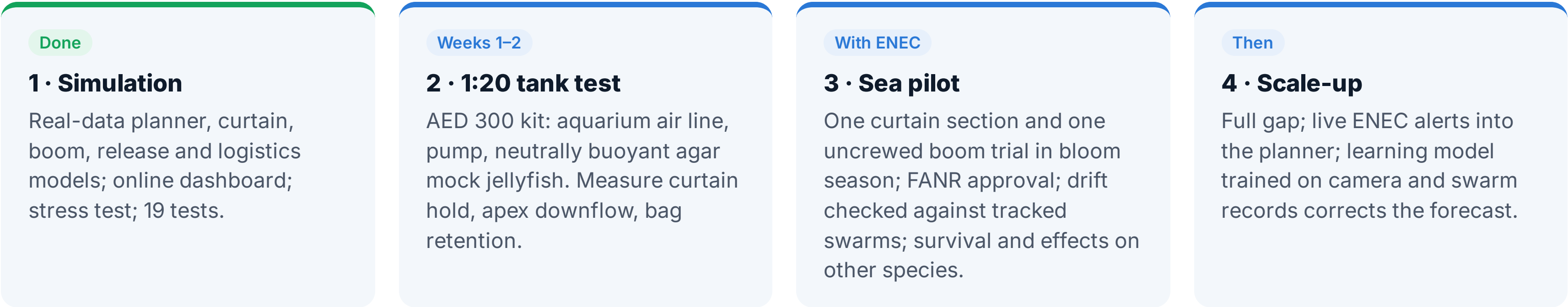
ASSUMPTION	BASIS	IF WRONG
Intake behind a breakwater with one gap, ~300 m (not yet measured), ~9.5 m deep	Brief; public channel depth	Curtain and compressor scale with width
Alert gives position, hours of notice	Brief: "time to prepare"	Camera trigger switches the curtain on
Swarm in top 1 m; adults swim $\leq 10\text{--}20$ cm/s	Field and lab studies (Australia)	Deeper skirt; tested at 20 cm/s
Swarm density 0.1–1 per m ³	No published figure	Capacity scales linearly
Apex downflow 25–50% of tow speed; bag passes 20%	Model hypothesis	Measured in the tank test

Limitations: the planner is tested only against simulations; currents come from one model point and the coast is a straight line; the curtain model ignores depth; ~8% of hours exceed the design airflow. **Why not harvest?** Blue Blubber is fished for food elsewhere; a full bag can go to a licensed processor instead of release. Release is the default while no UAE market exists.



Sensitivity: approach current, bag leakage / apex downflow and gap width drive the results, so we measure them first.

From simulation to sea, one measured step at a time

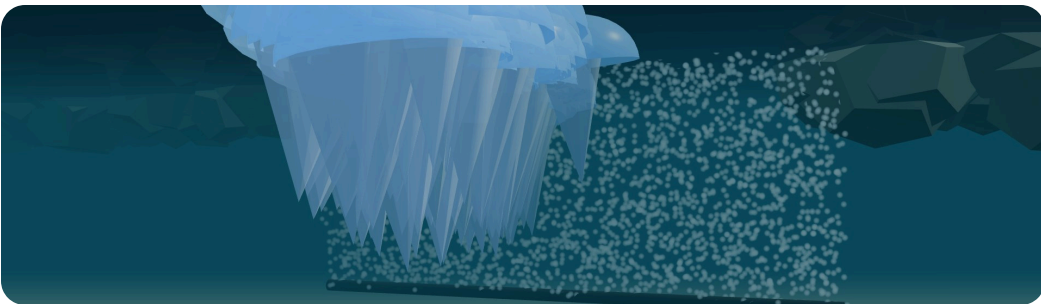
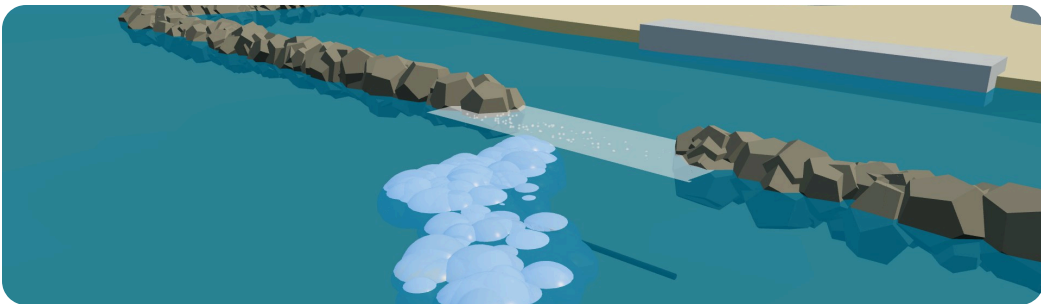


Pass marks

- Tank: $\geq 95\%$ held at the scaled 0.2 m/s current
- Tank: bag keeps $\geq 90\%$ at the scaled 0.2 m/s tow
- Pilot: no harm to non-target species; fewer screen-clearing hours

Open work and AI use

Live dashboard: enechackathon.streamlit.app. Code, data and tests: github.com/Shaaha-7/ENEC. AI assistants helped search and summarise literature, check claims, draft text and write simulation code, and an open text-to-speech model voices the video; the team reviewed every decision, assumption and figure, and can explain each in our own words.



Stills from our 3D explainer video: the curtain holding a swarm outside the gap, and the bubble wall seen underwater.